

Kernel Trick - Reducing Computational Costs

Let us say we have 2 points X & Y $X = (2, 3, 4)$
 $Y = (3, 4, 5)$

$$K(X, Y) = \langle f(X) \cdot f(Y) \rangle$$

$\rightarrow K$ is the Kernel Function

* X & Y are the dimensional inputs

* f is the map from n -dimensions to m -dimensions

$\langle X, Y \rangle$ dot product

$$f(X) \cdot f(Y)$$

$$f(X) = X \cdot X^T$$

$$f(Y) = Y \cdot Y^T$$

$$X = (2, 3, 4) \quad Y = (3, 4, 5)$$

$$X \cdot X^T = \begin{pmatrix} 2 & 3 & 4 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \times 2 & 2 \times 3 & 2 \times 4 \\ 3 \times 2 & 3 \times 3 & 3 \times 4 \\ 4 \times 2 & 4 \times 3 & 4 \times 4 \end{pmatrix} = \begin{pmatrix} 4 & 6 & 8 \\ 6 & 9 & 12 \\ 8 & 12 & 16 \end{pmatrix}$$

$$f(X) = \begin{matrix} X_1 X_1 & X_1 X_2 & X_1 X_3 \\ X_2 X_1 & X_2 X_2 & X_2 X_3 \\ X_3 X_1 & X_3 X_2 & X_3 X_3 \end{matrix}$$

$$f(X) = \begin{pmatrix} 2 \times 2 & 2 \times 3 & 2 \times 4 \\ 3 \times 2 & 3 \times 3 & 3 \times 4 \\ 4 \times 2 & 4 \times 3 & 4 \times 4 \end{pmatrix} = \begin{pmatrix} 4 & 6 & 8 \\ 6 & 9 & 12 \\ 8 & 12 & 16 \end{pmatrix}$$

$$f(Y) = \begin{pmatrix} 9 & 12 & 15 \\ 12 & 16 & 20 \\ 15 & 20 & 25 \end{pmatrix}$$

$$f(X) \cdot f(Y) = \begin{pmatrix} 4 & 6 & 8 & 6 & 9 & 12 & 8 & 12 & 16 \\ 15 & 12 & 16 & 20 & 15 & 20 & 25 \end{pmatrix}$$

$$= 36 + 72 + 120 + 72 + 144 + 240 + 120 + 240 + 400 = 1444$$

$$K(X, Y) = (2 \times 3 + 3 \times 4 + 4 \times 5)^2 = (6 + 12 + 20)^2 = 38^2 = 1444$$